

Main Components of a Computer

1. Hardware

Definition

Hardware refers to the physical components of a computer that can be seen and touched.

Think:

Hardware = Physical parts

Examples:

- CPU
- Monitor
- Keyboard
- Mouse
- RAM
- Hard disk / SSD
- Motherboard
- Printer
- Speakers

For example, when you look at your computer:

Monitor	→ Hardware
Keyboard	→ Hardware
Mouse	→ Hardware
CPU	→ Hardware
SSD	→ Hardware
RAM	→ Hardware

You can physically touch these components.

2. Why Is Hardware Necessary?

Software cannot run without hardware.

Suppose you have a Python program:

```
print("Hello")
```

The Python program is **software**.

But where does it execute?

It executes using the computer's:

CPU

RAM

Storage

Motherboard

So software needs hardware to actually execute.

3. Software

Definition

Software is a collection of programs and instructions that tells computer hardware what to do.

Think:

Software = Instructions/programs

Examples:

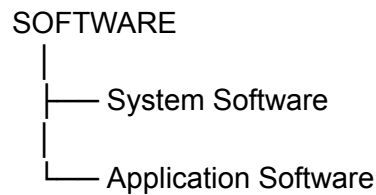
- Windows
- Linux
- Android
- Microsoft Word
- Google Chrome
- VS Code
- Python
- Games

You cannot physically touch the program itself.

You can touch the **storage device containing the program**, but you cannot touch the software as a physical object.

4. Two Main Types of Software

Your slide divides software into:



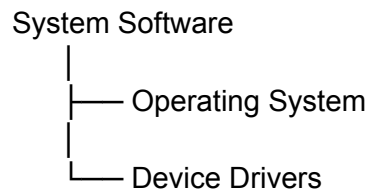
Let's understand both.

5. System Software

Definition

System software manages and controls the computer hardware and provides a platform for application software to run.

Your slide gives two examples:



A. Operating System

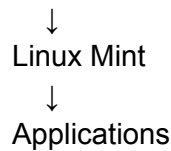
An **Operating System (OS)** is the main software that manages the computer's resources and provides an environment for applications.

Examples:

Windows
Linux
macOS
Android
iOS

Since you're using Linux Mint, for example:

Hardware

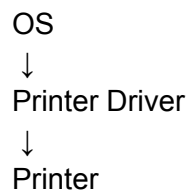


Linux Mint communicates with and manages the underlying hardware while providing the environment in which your applications run.

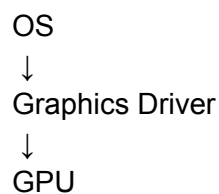
B. Device Drivers

A **device driver** is software that allows the operating system to communicate with and control a particular hardware device.

For example:



Or:



Without the appropriate driver, an operating system may not be able to properly use a particular device.

Simple way to remember:

Driver = Translator between the OS and a hardware device.

6. Application Software

Definition

Application software is software designed to help the user perform specific tasks.

Examples:

Microsoft Word → Writing
Chrome → Web browsing
VS Code → Programming
Calculator → Calculations
Photoshop → Image editing
Games → Entertainment

These programs are designed primarily for **user tasks**.

7. System Software vs Application Software

This distinction is very important for exams.

System Software	Application Software
Manages computer resources	Performs user-oriented tasks
Works closely with hardware	Usually works through the OS
Provides a platform for applications	Runs on that platform
Examples: Operating systems, drivers	Examples: Word, Chrome, VS Code

Easy memory trick:

System software → runs/manages the computer

Application software → helps the user do a task

8. Hardware + Software Work Together

Neither one is enough by itself.

Imagine:

HARDWARE
+
SOFTWARE
↓

COMPUTER SYSTEM

Example: Opening a browser

You click the Chrome icon.

Your:

Mouse → Hardware

provides input.

The operating system:

Software

handles the interaction.

The CPU:

Hardware

executes instructions.

RAM:

Hardware

holds information needed during execution.

The display:

Hardware

shows the browser window.

The browser:

Software

performs the web-browsing task.

So the complete process involves **both hardware and software**.

9. A Very Important Analogy

Think about a **car**.

Car body
Engine
Wheels
Brakes
Steering

These are physical → **Hardware**

Now imagine instructions telling the driver how to operate the car.

The instructions aren't physical parts of the car → similar idea to **Software**.

Another common analogy is:

Hardware is the body; software provides the instructions.

But remember, this is only an analogy. A computer is more complex than this comparison.